

Long term effects of childhood exposure to violence in fragile and conflict affected settings

Zulfiqar A Bhutta and colleagues argue that the response to children in fragile and conflict affected settings must move beyond event based framings to tackle cumulative and intergenerational adversity and contextually relevant opportunities for action

Early exposure to armed conflict is a growing global crisis, with more than 460 million children living in armed conflict zones.¹ Children in fragile and conflict affected settings face sustained exposure to violence, poverty, insecurity, family disruption, and service deprivation during sensitive periods of brain and psychosocial development, placing their mental health at heightened risk. This article builds on the work on health and nutrition in conflict settings done by BRANCH (Bridging Research and Action in Conflict Settings for the Health of Women and Children) Consortium. As part of the *BMJ* Collection on Child Mental Health in Conflict Settings, we describe how global responses fail to address the pathways through which early adversity in such settings affects children's mental health and development across the life course and generations. Other articles in the collection identify evidence based interventions and integration strategies to improve child outcomes and strengthen mental health systems.^{2,3}

KEY MESSAGES

- Conflict dismantles the familial, social, institutional, and environmental structures essential to child wellbeing
- This leads to cascading disruptions in brain development, stress physiology, attachment systems, and environmental learning
- Emerging epigenetic research suggests that trauma can be transmitted intergenerationally, shaped by maternal mental health, the caregiving environment, social conditions, and developmental timing
- Dominant global response models and funding often favour short term interventions tailored to donor reporting cycles rather than the required longer term, system embedded care

Inadequacy of current humanitarian responses

Fragile and conflict affected settings increasingly exemplify the global poly-crisis, in which disruptions across global systems become causally entangled and mutually reinforcing. Armed conflict coincides with extreme weather events amplified by climate change, pandemic aftershocks, widespread societal and political polarisation, greater economic uncertainty, rising child malnutrition, and more. In these settings, children's young age co-exists with other vulnerability factors such as poverty, displacement, disruptions in education and health services, and malnutrition, ultimately multiplying their risk for poor mental health and development outcomes.^{4,5} Humanitarian need has doubled in the past five years, with children representing half of those in need (90.3 million). More children are in need now than at any point since the second world war.⁴ However, global budgets for humanitarian aid have declined, deepening the risk in fragile and conflict affected settings.⁶

A large share of the behavioural and implementation evidence that underpins aid is built on western populations, limiting external validity and highlighting the need for globally representative samples and researchers. Experts warn against exporting individualised, event focused templates to non-western contexts, where interdependent agency, kinship obligations, and collective coping dominate.⁷ Child focused guidance itself notes that early childhood development framing is deeply embedded in western values, with evidence from high income countries applied uncritically to low and middle income countries.⁶

Additionally, humanitarian evidence hierarchies often favour short, standardised trials with easily measurable outcomes. Thus, practice and funding tend to favour brief, modular interventions suited to randomised controlled trials and donor reporting cycles. The result is limited external validity and cultural fit, emphasis on proximal and time bound outcomes,

and fragmentation driven by short project grants and output metrics rather than multiyear, system embedded care. A funding evaluation in 2017-21 using data from the United Nations Financial Tracking System estimated that only 3% of humanitarian funding for mental health and psychosocial support reached local and national agencies, constraining cultural adaptation and continuity of care.⁸ Beyond finance, a conceptual/empirical analysis in 2025 shows how standard capacity strengthening and participation schemes are framed in ways that maintain the status quo, limiting equal decision making with local partners.⁹ Decolonising reviews argue that risk instruments and funding calls impose high income country standards that exclude local actors, leading to stagnant localisation despite policy commitments. Taken together, these patterns explain why dominant responses remain fragmented, short term, and externally defined though children's mental health and development are cumulative and intergenerational.⁴

Effects of early adversity on children

Chronic stress alters brain architecture

Instead of experiencing several distinct sources of stress, young people living in fragile and conflict affected settings experience interconnected, chronic, and compounding stress associated with direct exposure to traumatic events, displacement, poverty, family separation, and disrupted services and care giving environments.⁵ As a result, children experience post-traumatic stress, anxiety, and depressive symptoms, which often co-occur and may persist into adolescence and adulthood. Ongoing instability limits access to protective factors such as consistent care giving, education, and mental health services, thereby constraining recovery and increasing the risk of adverse developmental trajectories.¹

Disruptions during critical windows can have disproportionately large and irreversible effects on developmental

trajectories. Lupien and colleagues reviewed the effects of stress across the life course and found that exposure in the prenatal period is linked to learning impairments, enhanced sensitivity to substance abuse, and increased anxiety and depression related behaviours in adulthood.¹⁰ During early childhood, especially prenatal to age five, the brain undergoes rapid growth and synaptic formation. Experiences during this critical window heavily influence the development of core neural systems that regulate stress physiology, language, emotional regulation, executive function, and attachment behaviour.⁵ Lupien and colleagues found that postnatal stress induces sustained and frequent activation of hormones and neurochemical systems, including the sympathetic-adrenomedullary and hypothalamic-pituitary-adrenocortical systems, which control cortisol and adrenaline production.¹⁰ Studies show that long term rises in cortisol levels can alter the function of several neural systems, suppress the immune response, and alter brain architecture in regions essential for learning and memory. For instance, a longitudinal observational study conducted among Syrian refugee children (year 1: $n=1574$; year 2: $n=923$) living in informal tented settlements in Lebanon found that greater exposure to war events and current symptoms of post-traumatic stress disorder predicted higher hair cortisol concentrations, which indicates hypersecretion of cortisol and long term changes to the hypothalamic-pituitary-adrenocortical axis following war related events and ongoing stressors of displacement.¹¹ Adolescence represents another critical window. Ongoing brain maturation in the prefrontal cortex and limbic system and heightened hypothalamic-pituitary-adrenocortical reactivity acutely sensitise adolescents to trauma exposure and stress related disorders.¹⁰ Prevailing evidence indicates that individual therapeutic interventions are unlikely to improve outcomes while children remain in unsafe or abusive environments; however, this principle has not been adequately applied in fragile and conflict affected settings, where children are unable to escape ongoing exposure to war related trauma.¹

Impacts persist and accumulate across generations

A review of human and animal evidence found that trauma related risk can be passed to offspring via epigenetic path-

ways.¹² The authors highlight two routes: preconception changes in parents that may alter offspring stress response and psychopathology risk; and developmental programming through early environments, including in utero maternal stress and postnatal caregiving quality. They conclude that current findings are suggestive but not definitive, calling for prospective, multigenerational studies integrating biology with social context to clarify mechanisms and inform prevention.¹²

Recent research elucidates how parental characteristics shape offspring DNA methylation profiles. Differentially methylated regions (eg, in the MTNR1B gene) associated with maternal age suggest that parental factors influence the epigenome of offspring at birth, including effects on crucial pathways linked to stress and resilience. Accelerated epigenetic ageing is induced prenatally in mothers exposed to violence, which implies that the prenatal environment is a critical window during which epigenetic programming may occur. Early life exposure, whether to psychosocial stress, trauma, or biological variables, can induce lasting epigenetic changes.¹³ A recent meta-analysis expands the scope to include paternal mental distress in the perinatal period as a predictor of child developmental outcomes and emphasises that epigenetic and behaviour outcomes are not solely maternal in origin. Paternal depression, anxiety, and stress are associated with poorer cognitive, language, and social-emotional development in children through both behavioural interactions and possibly germline epigenetic modifications.¹⁴

The field is converging on a model where early adversity and parental contexts jointly shape a child's epigenetic landscape. The intergenerational transmission of trauma can perpetuate cycles of psychological distress and health disparities, and dealing with these effects requires long term, adaptive interventions that operate at individual, familial, and societal levels, and extend well beyond the immediate aftermath of war.

Violent and disrupted care giving environments

In fragile and conflict affected settings, children's mental health and development are powerfully shaped by caregiver-child pathways in which organised violence spills into the home, degrades caregiver wellbeing, and transmits risk to children.⁵ Among Sri Lankan youth, the combined

impact of both the protracted civil war and a devastating tsunami on psychological functioning was significantly mediated by daily stressors such as deprivation, inter-parental conflict, and child abuse. Importantly, deprivation and child abuse were better predictors of post-traumatic stress disorder than direct exposure to disaster or conflict, underscoring how layered, chronic stress drives disorder risk.¹⁵ Moreover, war and displacement profoundly affect caregiving environments, with adverse consequences for child mental health. Family level stressors include the loss of caregivers, prolonged separation, parental mental health difficulties, and unemployment. Unaccompanied minors are particularly vulnerable to trauma through sexual or physical exploitation, abuses in institutional care, and illegal labour.¹

Too often, research on children in non-western contexts overlooks resilience mechanisms and the protective factors that sustain them. Neuroscientific studies reveal that caregiving environments that provide emotional support, responsive parenting, and opportunities for exploration buffer the effects of trauma and poverty on brain development. Synchrony, the coordinated biological and behavioural exchange between caregivers and children, is foundational for emotional regulation, stress resilience, and social development. Synchrony also extends beyond dyadic relationships, enabling connections to abstract social structures such as communities or religious beliefs, which are critical in contexts of displacement.¹⁶ A qualitative study of Syrian refugees in Lebanon found that adolescent resilience was rooted in family, religious faith, and local networks including extended family. Coping drew on shared routines and community belonging rather than solely individual traits. Everyday routines, such as praying together five times a day, observing Ramadan, and maintaining faith based community structures, have a crucial role in fostering resilience.¹⁷

Service breakdown and material deprivation transmit harm across systems

Conflict and deprivation consistently impair health, learning, and psychosocial wellbeing through pathways including service disruption, attacks on schools, and nutritional deficits.¹⁸ Ongoing civil war in Yemen, the genocide in the Middle East, more than a decade of war in Syria, the Sudanese civil war, ongoing hostilities in Afghanistan, and the Rohingya refugee

crisis degrade developmental contexts. Children's main sources of protection and safety (eg, family, peer networks, community systems) are disrupted. Social protection and mental health coverage mitigate poor outcomes; however, evidence reports service disruptions in fragile and conflict affected settings, with downstream effects on immunisation, growth monitoring, and perinatal care. For instance, after years of plateau, global routine immunisation coverage fell from 86% in 2019 to 81% in 2021 amid covid-19 related disruptions, which contributed to large measles outbreaks.⁴ Routine immunisation is a key determinant of child development through its impact on morbidity and school readiness. Overall, health system disruptions impair preventive services and thereby undermine foundational conditions for healthy child development.⁴

The Global Coalition to Protect Education from Attack reported about 6000 attacks or military incidents to have harmed over 10000 students and educators in 2022-23, with the Middle East and North Africa and South Asia having among the most attacks on education.¹⁹ Moreover, before the pandemic, about 15 million children in the Middle East and North Africa region were out of school and an additional 10 million were at risk of leaving school because of poverty, marginalisation, displacement, and disruption caused by armed conflict.¹⁸ These conditions are closely linked to long term learning impairment and worsened mental health, particularly among disadvantaged families. The loss of safe, predictable environments and daily routine prevents exploratory and play based learning essential for healthy development.¹⁹

The Global Report on Food Crises in 2025 found that conflict and insecurity

are the primary drivers of acute food insecurity and child malnutrition, with 2024 having record levels and severe hotspots across Gaza, Yemen, and Sudan and spillovers into Afghanistan and Pakistan.²⁰ Additionally, they reported record levels of child acute malnutrition where access to food is deliberately restricted as a method of warfare, though the tactic is prohibited by the Geneva Conventions.²⁰ Conflict and concurrent crises destabilise food systems and household income, driving malnutrition with long term cognitive and mental health consequences.¹⁸ Evidence from low and middle income countries suggests that early nutrition deficiency is linked to developmental losses, including poor long term brain function, cognition, and motor and socioemotional development.¹⁵

Counterviews

Some people argue that critics of western based published work overlook both universal features of human psychology and individual distinctions (eg, health sequelae, early temperaments) that shape outcomes. Our argument does not deny universal or individual factors but calls for empirical acknowledgment of population variability across geographies and cautions against generalising outlier samples.⁷

We do not suggest converting all acute actions into long term programmes. Acute phase responses are key to saving lives, removing children and families from immediate harm, and creating platforms for intervention and sustained care. Clinical task sharing packages (eg, the Mental Health Gap Action Programme) and brief, scalable psychotherapies are supported by evidence from randomised and pragmatic trials in populations exposed to adversity, relieving symptoms

and creating demand for longer term support.²¹ Moreover, the field is not uniformly short term oriented. Core guidance and recent reviews call for layered, life course, and systems approaches that many countries are adopting. However, the incentive structure still skews towards the short term. Localisation faces real risk and implementation constraints that bias systems towards short projects and large intermediaries. However, recent "risk sharing" work argues that localisation can be improved when donors and international non-governmental organisation partners share fiduciary and legal risk rather than shifting the burden to local actors.²²

Finally, evidence is limited by measurement and generalisability issues. Much of the field relies on subjective self-reported and parent reported data and cross sectional designs. Measurement instruments built in western contexts may not map on to local meanings, which may mischaracterise, underdetect, or pathologise local experiences of distress, wellbeing, and resilience.

Improving global responses

Prevailing global responses treat conflict as a sequence of discrete events amenable to fragmented, short term interventions. However, children living in fragile and conflict affected settings face interacting crises, including pandemic aftershocks, cost-of-living pressures, climate hazards, and service breakdowns that compound over time. Concurrent crises threaten children's mental health and development through inter-related mechanisms that alter brain architecture and stress response systems, leading to increased risk of stress related disorders, impaired learning, emotional dysregulation, and more. Moreover, mental health and development consequences may be biologically embedded and psychosocially transmitted from caregivers to their children, suggesting that concurrent crises may reshape neurodevelopmental, emotional, and relational trajectories across generations. Box 1 sets out recommendations for a model of care that acknowledges these effects. Improved understanding of the biology of early childhood stress and mental health consequences in fragile and conflict affected settings allows us to develop interventions that work across a range of contexts.

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Box 1: Recommendations for supporting children in fragile and conflict affected settings

A trajectory aware model that is attuned to toxic stress, caregiver pathways, deprivation, and intergenerational transmission during concurrent crises requires:

- Long cycle response programming that repositions short term tools as bridges into systems embedded care
- Family centred supports integrated with mental health and psychosocial support and social protection
- Continuity across platforms (eg, primary care, schools, social protection)
- Long term funding for systems transformation tied to the humanitarian-development-peace nexus and national systems
- Hybrid effectiveness-implementation and pragmatic trials, realist and mixed methods evaluations, longitudinal follow-up, and culturally anchored measures with study populations beyond western contexts
- Data tracking with longitudinal indicators (eg, embed mental health and psychosocial support and early childhood development indicators in health and education management information systems)

the paper. NS, who has worked on multiple global maternal and child health research projects, led the evidence synthesis and drafting of the manuscript. RD contributed expertise on transmission, resilience, and protective processes. FT, CB, and NT provided critical review and regional perspectives. Sources comprised peer reviewed systematic reviews and meta-analyses, Unicef and WHO guidance and data, and authoritative reports complemented by recent studies from the Middle East, North Africa, and South Asia. All authors approved the final manuscript and references for submission. The views are those of the authors and do not represent their affiliated organisations.

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